Security for Network Systems Lab 7

3.4 Capture and Analyze tshark Data

The student used tshark to monitor communication between the attacker and the victim machine. The capture showed a continuous flow of encrypted SSH packets between the relevant IP addresses. These encrypted packets indicate that the backdoor relies on secure communication channels that conceal their contents. Alongside the encrypted traffic, the capture included regular TCP acknowledgment packets confirming that the connection remained stable and reliable. The presence of Spanning Tree Protocol (STP) frames showed that the surrounding network was functioning normally, and ARP requests indicated that both systems were correctly resolving each other's MAC addresses.

Command Set 1 (TCP / SSH)

For the first command set, the communication behaved like standard encrypted SSH traffic transmitted over TCP. The connection was stable, with frequent acknowledgment packets confirming packet delivery. This reliable communication ensured the backdoor could execute commands in the correct order while remaining concealed due to the encryption.



Command Set 2 (UDP)

In the second command set, the traffic appeared consistent with UDP-based behaviour, where packets were sent without acknowledgment. This made the exchange faster but less reliable, as packet loss could occur. In a backdoor scenario, this could result in commands not being delivered or received consistently, making it less dependable than TCP.



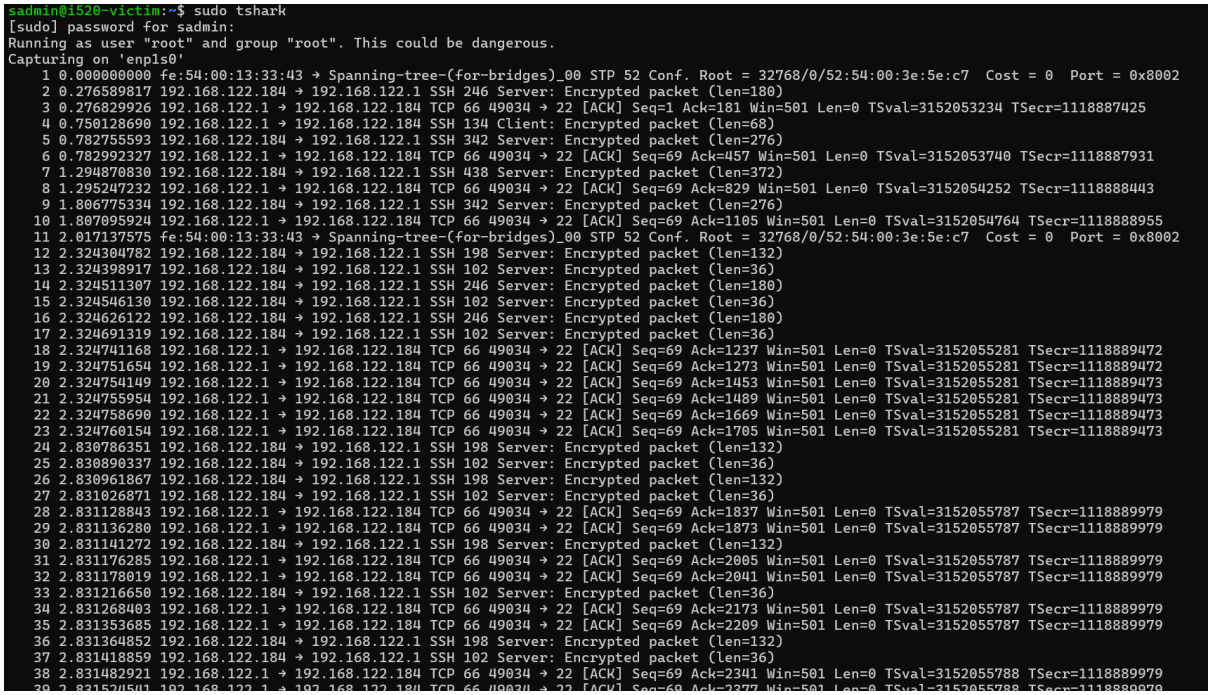
Command Set 3 (TCP)

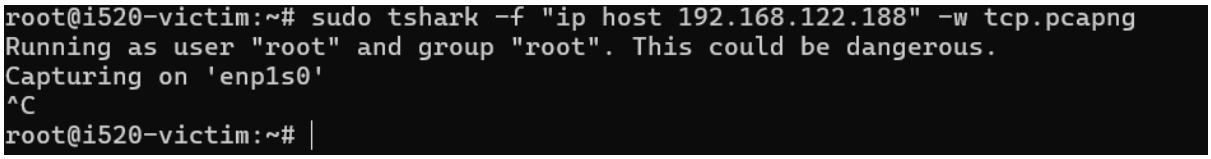
The third command set showed behaviour characteristic of TCP again, and in some cases resembled typical HTTP or HTTPS traffic patterns. This allowed the backdoor to blend seamlessly into normal web activity, making detection significantly more challenging for security tools or administrators reviewing network logs.



Comparison Summary

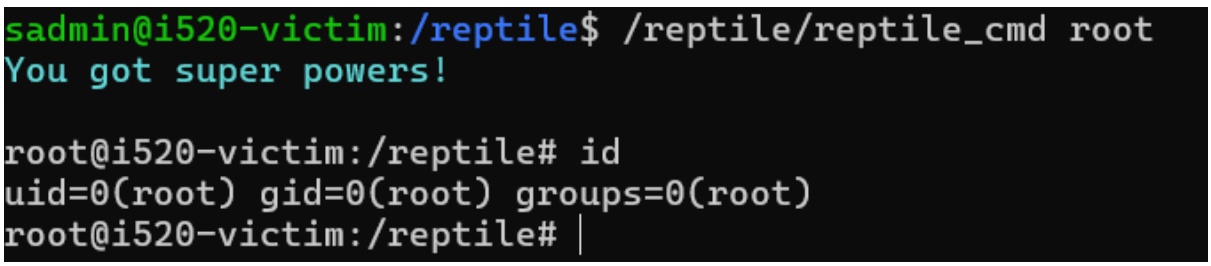
Across all three command sets, each protocol demonstrated different operational and stealth characteristics. TCP, used in the first and third sets, provided structured and reliable communication ideal for sustained backdoor control. UDP, used in the second set, allowed faster but unstable communication that could fail if packets were dropped. When the backdoor traffic resembled HTTP or HTTPS behavior, it became highly disguised within ordinary network traffic, reducing the likelihood of detection. The selection of ports also influenced stealth: standard ports such as SSH's port 22 might attract attention, whereas high ephemeral ports or common web ports like 80 or 443 helped the backdoor blend into legitimate traffic.





3.5 Assuming Root Privileges Without sudo

To verify that the Reptile rootkit enabled privilege escalation without using sudo, the victim machine was logged in as normal, unprivileged user. After triggering the backdoor with the configured parameters, the shell immediately switched to a root session without requesting the root password. This confirmed that the rootkit had successfully installed a covert privilege-escalation mechanism.



3.6 Rootkit Detection

To detect the presence of the rootkit, Tripwire was initially used to compare the system's current state against the baseline taken before the rootkit installation. Tripwire did not directly identify Reptile, but it did highlight unusual file changes that suggested

unauthorized system modification. Additional tools, including chkrootkit and rkhunter, generated warnings about hidden processes and suspicious kernel-related artifacts. These findings, supported by tshark packet captures showing encrypted backdoor communication, confirmed the installation of the rootkit.

```
sadmin@i520-victim:/$ sudo tripwire --check
[sudo] password for sadmin:
Parsing policy file: /usr/local/etc/tw.pol
*** Processing Unix File System ***
Performing integrity check...
The object: "/run" is on a different file system...ignoring.
The object: "/sys" is on a different file system...ignoring.
### Warning: File system error.
### Filename: /usr/local/libexec
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /usr/local/sysinfo
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /usr/libexec
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /usr/X11R6/lib
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /usr/tmp
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /etc/adjtime
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /etc/bashrc
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /etc/csh.cshrc
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /etc/csh.login
### No such file or directory
### Continuing...
### Warning: File system error.
### Filename: /etc/mail/statistics
```

```
17.  File system error.
     Filename: /var/lib/rpm/__db.003
     No such file or directory
18.  File system error.
     Filename: /var/lib/logrotate.status
     No such file or directory
19.  File system error.
     Filename: /var/lost+found
     No such file or directory
20.  File system error.
     Filename: /var/cache/man/whatis
     No such file or directory
21.  File system error.
     Filename: /cdrom
     No such file or directory
22.  File system error.
     Filename: /floppy
     No such file or directory
23.  File system error.
     Filename: /initrd
     No such file or directory
24.  File system error.
     Filename: /home/lost+found
     No such file or directory
25.  File system error.
     Filename: /root/.bash_profile
     No such file or directory
26.  File system error.
     Filename: /root/.cshrc
     No such file or directory
27.  File system error.
     Filename: /.autofsck
     No such file or directory

------------------------------------------------------------------------------
*** End of report ***

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for details use --version. This is free software which may be redistributed
or modified only under certain conditions; see COPYING for details.
All rights reserved.
Integrity check complete.
sadmin@i520-victim:/$
```

```
sadmin@i520-victim:/$ sudo chkrootkit
ROOTDIR is `/'
Checking `amd'...                                           not found
Checking `basename'...                                      not infected
Checking `biff'...                                          not found
Checking `chfn'...                                          not infected
Checking `chsh'...                                          not infected
Checking `cron'...                                          not infected
Checking `crontab'...                                       not infected
Checking `date'...                                          not infected
Checking `du'...                                            not infected
Checking `dirname'...                                       not infected
Checking `echo'...                                          not infected
Checking `egrep'...                                         not infected
Checking `env'...                                           not infected
Checking `find'...                                          not infected
Checking `fingerd'...                                       not found
Checking `gpm'...                                           not found
Checking `grep'...                                          not infected
Checking `hdparm'...                                        not infected
Checking `su'...                                            not infected
Checking `ifconfig'...                                      not infected
Checking `inetd'...                                         not infected
Checking `inetdconf'...                                     not found
Checking `identd'...                                        not found
Checking `init'...                                          not infected
Checking `killall'...                                       not infected
Checking `ldsopreload'...                                   not infected
Checking `login'...                                         not infected
Checking `ls'...                                            not infected
Checking `lsof'...                                          not infected
Checking `mail'...                                          not infected
Checking `mingetty'...                                      not found
Checking `netstat'...                                       not infected
Checking `named'...                                         not found
Checking `passwd'...                                        not infected
Checking `pidof'...                                         not infected
Checking `pop2'...                                          not found
Checking `pop3'...                                          not found
Checking `ps'...                                            not infected
Checking `pstree'...                                        not infected
Checking `rpcinfo'...                                       not found
Checking `rlogind'...                                       not found
Checking `rshd'...                                          not found
Checking `slogin'...                                        not infected
```

```
Searching for Suckit rootkit...                          nothing found
Searching for Volc rootkit...                            nothing found
Searching for Gold2 rootkit...                           nothing found
Searching for TC2 Worm default files and dirs...         nothing found
Searching for Anonoying rootkit default files and dirs...  nothing found
Searching for ZK rootkit default files and dirs...       nothing found
Searching for ShKit rootkit default files and dirs...    nothing found
Searching for AjaKit rootkit default files and dirs...   nothing found
Searching for zaRwT rootkit default files and dirs...    nothing found
Searching for Madalin rootkit default files...           nothing found
Searching for Fu rootkit default files...                nothing found
Searching for ESRK rootkit default files...              nothing found
Searching for rootedoor...                               nothing found
Searching for ENYELKM rootkit default files...           nothing found
Searching for common ssh-scanners default files...       nothing found
Searching for Linux/Ebury - Operation Windigo ssh...     not tested
Searching for 64-bit Linux Rootkit ...                   nothing found
Searching for 64-bit Linux Rootkit modules...            nothing found
Searching for Mumblehard Linux ...                       nothing found
Searching for Backdoor.Linux.Mokes.a ...                 nothing found
Searching for Malicious TinyDNS ...                      nothing found
Searching for Linux.Xor.DDoS ...                         nothing found
Searching for Linux.Proxy.1.0 ...                        nothing found
Searching for suspect PHP files...                       nothing found
Searching for anomalies in shell history files...        nothing found
Checking `asp'...                                        not infected
Checking `bindshell'...                                  not infected
Checking `lkm'...                                        chkproc: nothin
g detected
chkdirs: nothing detected
Checking `rexedcs'...                                    not found
Checking `sniffer'...                                    lo: not promisc
 and no packet sniffer sockets
enp1s0: PACKET SNIFFER(/lib/systemd/systemd-networkd[573])
Checking `w55808'...                                     not infected
Checking `wted'...                                       chkwtmp: nothin
g deleted
Checking `scalper'...                                    not infected
Checking `slapper'...                                    not infected
Checking `z2'...                                         chklastlog: not
hing deleted
Checking `chkutmp'...                                    chkutmp: nothin
g deleted
Checking `OSX_RSPLUG'...                                 not tested
sadmin@i520-victim:/$
```

```
sadmin@i520-victim:/$ sudo rkhunter --check
[ Rootkit Hunter version 1.4.6 ]

Checking system commands...

  Performing 'strings' command checks
    Checking 'strings' command                               [ OK ]

  Performing 'shared libraries' checks
    Checking for preloading variables                        [ None found ]
    Checking for preloaded libraries                         [ None found ]
    Checking LD_LIBRARY_PATH variable                        [ Not found ]

  Performing file properties checks
    Checking for prerequisites                               [ OK ]
    /usr/sbin/adduser                                        [ OK ]
    /usr/sbin/chroot                                         [ OK ]
    /usr/sbin/cron                                           [ OK ]
    /usr/sbin/groupadd                                       [ OK ]
    /usr/sbin/groupdel                                       [ OK ]
    /usr/sbin/groupmod                                       [ OK ]
    /usr/sbin/grpck                                          [ OK ]
    /usr/sbin/nologin                                        [ OK ]
    /usr/sbin/pwck                                           [ OK ]
    /usr/sbin/rsyslogd                                       [ OK ]
    /usr/sbin/sshd                                           [ OK ]
    /usr/sbin/useradd                                        [ OK ]
    /usr/sbin/userdel                                        [ OK ]
    /usr/sbin/usermod                                        [ OK ]
    /usr/sbin/vipw                                           [ OK ]
    /usr/sbin/unhide                                         [ OK ]
    /usr/sbin/unhide-linux                                   [ OK ]
    /usr/sbin/unhide-posix                                   [ OK ]
    /usr/sbin/unhide-tcp                                     [ OK ]
    /usr/bin/awk                                             [ OK ]
    /usr/bin/basename                                        [ OK ]
    /usr/bin/chattr                                          [ OK ]
    /usr/bin/curl                                            [ OK ]
    /usr/bin/cut                                             [ OK ]
    /usr/bin/diff                                            [ OK ]
    /usr/bin/dirname                                         [ OK ]
    /usr/bin/dpkg                                            [ OK ]
    /usr/bin/dpkg-query                                      [ OK ]
    /usr/bin/du                                              [ OK ]
    /usr/bin/env                                             [ OK ]
```

```
    Checking for passwd file changes                        [ None found ]
    Checking for group file changes                         [ None found ]
    Checking root account shell history files               [ OK ]

  Performing system configuration file checks
    Checking for an SSH configuration file                  [ Found ]
    Checking if SSH root access is allowed                  [ Warning ]
    Checking if SSH protocol v1 is allowed                  [ Not set ]
    Checking for other suspicious configuration settings    [ None found ]
    Checking for a running system logging daemon            [ Found ]
    Checking for a system logging configuration file        [ Found ]
    Checking if syslog remote logging is allowed            [ Not allowed
]

  Performing filesystem checks
    Checking /dev for suspicious file types                 [ None found ]
    Checking for hidden files and directories               [ None found ]

[Press <ENTER> to continue]



System checks summary
=====================

File properties checks...
    Files checked: 147
    Suspect files: 0

Rootkit checks...
    Rootkits checked : 500
    Possible rootkits: 2
    Rootkit names    : Diamorphine LKM, IntoXonia-NG Rootkit

Applications checks...
    All checks skipped

The system checks took: 1 minute and 52 seconds

All results have been written to the log file: /var/log/rkhunter.log

One or more warnings have been found while checking the system.
Please check the log file (/var/log/rkhunter.log)

sadmin@i520-victim:/$
```

```
sadmin@i520-victim:/$ sudo unhide quick
Unhide 20130526
Copyright © 2013 Yago Jesus & Patrick Gouin
License GPLv3+ : GNU GPL version 3 or later
http://www.unhide-forensics.info

NOTE : This version of unhide is for systems using Linux >= 2.6

Used options:
[*]Searching for Hidden processes through  comparison of results of system
calls, proc, dir and ps

sadmin@i520-victim:/$ sudo unhide.rb quick
Scanning for hidden processes...
No hidden processes found!
sadmin@i520-victim:/$
```

4.0 Submission

1. How did you determine the rootkit is installed? Did Tripwire detect it? What confirmed the infection?

Tripwire is used to compare the current system state with its original baseline. Although Tripwire did not explicitly name the Reptile rootkit, it flagged several unauthorized file modifications that raised suspicion. Tshark captured continuous encrypted communication between the attacker and the victim, suggesting the presence of a hidden channel. Tools such as chkrootkit and rkhunter provided clearer confirmation, as they reported hidden processes and suspicious kernel behavior consistent with known rootkits. These combined results verified that the rootkit had been installed.

2. Where is the rootkit installed in the victim VM?

The Reptile rootkit was installed in the directory /root/Reptile/sbin on the victim machine. This location contained the installation scripts, the kernel module, and support components used by the backdoor.

3. If detection tools were not used, how could you still find the rootkit?

Even without automated detection tools, the rootkit could be discovered through manual analysis. The student could inspect active processes using ps aux, top, or pstree to look for unexpected activity. Hidden files could be uncovered with recursive directory listings using ls -la and find /. Examining loaded kernel modules with lsmod, /proc/modules, or modinfo would reveal anomalies. Network inspection commands such as netstat, ss, or lsof -i could identify unknown listening ports or unauthorized external connections. System logs, including /var/log/auth.log, syslog, and dmesg, would also provide hints of unusual behavior. These checks would allow detection even without specialized tools.

4. Can you disable the backdoor without uninstalling the rootkit?

Yes. The student can temporarily disable the backdoor by blocking the ports it uses through firewall rules, preventing incoming or outgoing traffic on the backdoor channels. The associated processes can also be terminated using commands such as kill or pkill. If the malicious kernel module is visible, unloading it with rmmod may also interrupt the backdoor's operation. However, doing so does not remove the rootkit entirely, and the backdoor may reactivate after reboot or through hidden persistence mechanisms.

5. How can you completely disable the rootkit?

To fully disable the rootkit, it must be entirely removed from the system. This can be done by executing Reptile's uninstall option if provided, or by manually deleting all rootkit files, removing the kernel module, and clearing any persistence entries. After removal, the system should be scanned again using tools like Tripwire, chkrootkit, or rkhunter to ensure no remnants remain. In cases where certainty is required, performing a full OS reinstallation after backing up essential data is the most reliable method to completely eliminate the rootkit.